

7

Vectors

Learning Objectives

In this chapter, we will

- perform addition and subtraction of vectors, multiplication of a vector by a scalar, and give geometrical interpretations of these vectors
- use vector notations such as $\begin{pmatrix} x \\ y \end{pmatrix}$, $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\mathbf{p} = x\mathbf{i} + y\mathbf{j}$, $\mathbf{q} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $\overline{AB}$ and $\mathbf{a}$
- distinguish between position vectors and displacement vectors
- calculate the magnitude of a vector
- find the unit vector of a given vector
- calculate the distance between two points as well as the angles between a vector and the x -, y - or z -axis
- apply ratio theorem in geometrical applications
- apply concepts of scalar product and vector product of vectors to solve problems involving vectors
- calculate the angle between two vectors using scalar product of vectors and the area of triangle or parallelogram using vector product of vectors
- give geometrical interpretations of $|\mathbf{a} \cdot \mathbf{b}|$ and $|\mathbf{a} \times \mathbf{b}|$, where $\mathbf{b}$ is a unit vector
- find vector and cartesian equations of lines and planes
- calculate the distance from a point to a line or to a plane
- find the foot of perpendicular from a point to a line or to a plane
- calculate the angle between two lines, between a line and a plane, or between two planes
- interpret geometrical relationships between
 - two lines (coplanar or skew)
 - a line and a plane
 - two planes
- find the intersections of lines and planes

§ 7.1 Introduction



All physical quantities can be classified as scalars or vectors.

Scalars are quantities which have magnitude but no associated direction. Hence, a scalar can be represented entirely by a real number.

Examples of scalars: length, mass, speed, volume, temperature, time.

Vectors are quantities which have magnitude as well as an associated direction.

Examples of vectors: displacement, weight, velocity, acceleration, force.

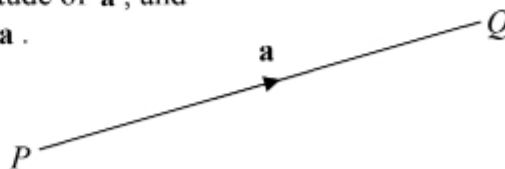
In this chapter, we will learn more about vectors and its geometrical representations in two dimensions and three dimensions.

▶ § 7.2 Representation of Vectors

A vector quantity is usually denoted by a letter in bold lowercase, e.g. **a**, **k**, **p**, in print. In written work, a vector quantity can be represented by a letter in lowercase with a tilde below the letter, e.g. $\underline{a}$, $\underline{k}$, $\underline{p}$.

Geometrically, a vector **a** can be represented by a directed line segment joining two points, say P and Q . In this case, the vector **a** can also be denoted by $\overrightarrow{PQ}$, where

- (i) the length of line segment PQ is equal to the magnitude of **a**, and
- (ii) the direction from P to Q indicates the direction of **a**.



🔍 Note

For written work, we shall use $\underline{a}$ or $\overrightarrow{PQ}$ to represent the vector **a**.

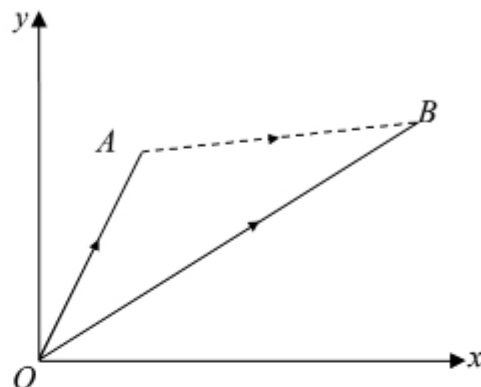
▶ Position and Displacement Vectors

A position vector represents the position of a point in space relative to the origin O . For example, the vector $\overrightarrow{OA}$ is known as the position vector of A with respect to O .

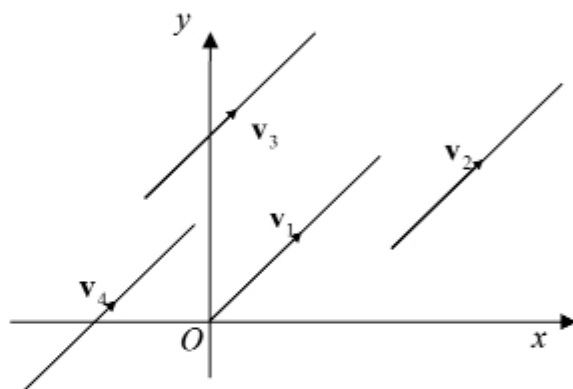
Given that A and B are any two points in the Cartesian plane. The vector $\overrightarrow{AB}$ can be found using the position vectors of A and B , where

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

This is a very useful result in finding vectors when their position vectors are known.



On the other hand, a displacement vector represents the length and direction of the imaginary straight path, i.e. the shortest distance, from an initial point to the final point. A displacement vector has no fixed starting point, and hence no fixed location. For example, if we are given a vector $\mathbf{v}$, we do not know the position of $\mathbf{v}$. In the diagram on the right, $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, and $\mathbf{v}_4$ are equal vectors with the same magnitude and direction.



In other words, if the vector $\overrightarrow{AB}$ is given, we do not know where the points A and B are. What we know is the relative position of B with respect to A .

